



General Sir John Kotelawala Defence University

Department of Computer Science

CS11021– Programming Laboratory (Intake 43)

Lab Sheet 3

1. Type a C++ program to display the following exactly as shown:

```
Welcome to C++ Programming!  
Have fun learning!
```

2. Write a program to display your name surrounded by stars. For example, if your name is KAUSHALYA, output should be:

```
*****  
* KAUSHALYA *  
*****
```

3. Write a program to display the following number pattern:

```
1  
12  
123  
1234  
12345
```

4. Write a program to display this smiley face using only cout:



5. Write a C++ program to display a restaurant bill exactly like this:

***** RESTAURANT BILL *****		
Item	Quantity	Price

Burger	2	\$5
Fries	1	\$2
Coke	3	\$1

TOTAL		\$12
Thank you for visiting!		

6. Write a C++ program to display a cricket match score table:

***** MATCH SCORE *****			
Team	Runs	Wickets	Overs

Team A	250	8	50
Team B	240	10	48.3

Winner: Team A			

7. Write a program to print a simple weekly calendar for a week:

Sun	Mon	Tue	Wed	Thu	Fri	Sat

1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

8. You purchased **5 apples**, and the price of each apple is **20 rupees**. Type a C++ program that calculates and displays the **total cost of the apples**.
9. A car travels at a **constant speed of 60 km/h** for **2.5 hours**. Type a C++ program that calculates and displays the **distance traveled** by the car.
10. Your household consumed **120 units of electricity**. The cost per unit is **15.5 rupees**, and there is an additional **fixed charge of 50 rupees**. Type a C++ program that calculates and displays the **total electricity bill**.
11. A temperature of **30°C** needs to be converted to Fahrenheit using the formula:

$$\text{Fahrenheit} = (\text{Celsius} \times \frac{9}{5}) + 32$$

Type a C++ program that converts Celsius to Fahrenheit and displays the result.

12. A worker works **40 hours** at **250 Rs/hour** and **5 overtime hours** at **300 Rs/hour**. Type a C++ program that calculates and displays the **total salary**, including overtime.

13. You are planning a road trip. Your car consumes **liters of fuel per 100 km**, and the fuel price is given. You need to calculate the **total fuel cost** for a trip.

Task:

- Type a C++ program that asks the user to **enter**:
Distance of the trip in kilometers (integer or float)
Car's fuel consumption in liters per 100 km (float)
Fuel price per liter (float)
- Calculate the total fuel cost using the formula:

$$\text{Total Cost} = \frac{\text{Distance} \times \text{Fuel Consumption}}{100} \times \text{Fuel Price}$$

- Display the **total fuel cost**.
- Use proper **variables, data types, and type casting** if needed.

14. A student has marks from **5 subjects**. The total marks for each subject is **100**. You need to calculate the **percentage score**.

Task:

- Type a C++ program that asks the user to **enter the marks obtained in 5 subjects**.
- Calculate the **total marks obtained** and the **percentage** using the formula:

$$\text{Percentage} = \frac{\text{Total Marks Obtained}}{500} \times 100$$

- Display the total marks obtained and the percentage.
- Use appropriate variables, data types, and type casting to ensure the percentage is displayed with decimals.

15. You want to calculate how much it will cost to recharge your mobile for a certain number of **days**, based on the **daily data usage** and **cost per GB of data**.

Task:

- Type a C++ program that asks the user to enter:
- Number of days they want to recharge for
- Daily data usage in GB
- Cost per GB of data
- Use these values to calculate and display the **total recharge cost**.